Fan Huang

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Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington

Education

2023–* **Ph.D.** in Informatics (Complex Networks and Systems), Indiana University Bloomington, USA
Advisors: Prof. Jisun An and Prof. Haewoon Kwak

2022–23 **Ph.D.** in Computer Science, Singapore Management University, Singapore
Advisors: Prof. Jisun An and Prof. Haewoon Kwak

2020–22 **M.Sc.** in Information Systems, Nanyang Technological University, Singapore
Advisor: Prof. Jin-Cheon Na
Thesis: *Public Opinion Trend Analysis on Nuclear Energy via Aspect-based Sentiment Analysis Methods*

2014–18 **B.Sc.** in Computer Science & Technology, Central South University, China
Advisor: Prof. Zuping Zhang

Platforms & Tools

AgentSims: Agentic Simulations agentsims.agentsnet.org

LLM-driven multi-agent simulation framework with persistent state and replayable histories for reproducible academic research on collective behavior. Designed to support industry scenario analysis, strategy evaluation, and risk assessment through configurable agent environments.

Founders-Sim: Market Competition Simulation founders-sim.agentsnet.org

Rule-based simulation of startups vying for users in a synthetic population of 80,000 groups (100,000 people each). Social ties spread word of products, but only advertising or referrals can start free trials, after which some pay. The site replays a recorded 1,000-week run.

Cytolume: GPU-Accelerated Million-Cell Visualization muyuhuatang.github.io/Cytolume

WebGL2 visualization library for large single-cell embeddings: UMAP, t-SNE, and PCA layouts in 2D and 3D, lasso selection, Human Reference Atlas anatomy beside the cell map, and figure export with data provenance. Successor to Human Reference Atlas cell-embedding notebooks built with Prof. Katy Börner's group in 2024. A Python package serves the viewer from AnnData objects, locally or in Jupyter.

*Expected March 2027.

Auto-Research Monitor muyuhuatang.github.io/auto-research-monitor

Live monitor for autonomous research agents: every question, experiment, run, result, and claim joins an evidence lineage graph, and each claim is scored green, yellow, or red on five structural dimensions of its supporting evidence, such as coverage, reproducibility, and data provenance. Works alongside any agent, such as Claude Code, via an HTTP API, a Python SDK, or a watched project folder, and can replay each run step by step.

Founders-Sim: Multiplayer Startup Race Game founders-game.agentsnet.org

Race between startups on a new 3D planet each round, populated by 80,000 personas whose wealth and chance of joining are drawn at random. Players hire a team and raise funds; when two startups meet, the stronger acquires the other, yet fog of war hides every rival's strength.

Publications

G Google Scholar Citations: 649 h-index: 6 (as of Oct. 2026)
† → Equal contribution

Peer-reviewed Conference Proceedings

C1. Yang, K., **Huang, Fan**, Chen, X. & He, J. *GaitEval: A Multi-Condition Benchmark for Evaluating Large Language Models in Clinical Gait Reasoning* in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (2027).

C2. **Huang, Fan**. *LLM Belief Atlas* in *InfoVis (Poster)* (2026).

C3. **Huang, Fan**, Kwak, H. & An, J. *Understanding Moral Reasoning Trajectories in Large Language Models: Toward Probing-Based Explainability* in *Findings of the Association for Computational Linguistics (EMNLP Findings)* (2026).

C4. **Huang, Fan**, Kwak, H. & An, J. *Vulnerability of LLMs' Stated Belief? LLMs Belief Resistance Check Through Strategic Persuasive Conversation Interventions* in *Findings of the Association for Computational Linguistics (ACL Findings)* (2026).

C5. Qi, W., **Huang, Fan**, Muralidharan, R., An, J. & Kwak, H. *XChoice: Explainable Evaluation of AI-Human Alignment in LLM-based Constrained Choice Decision Making* in *Findings of the Association for Computational Linguistics (AACL-IJCNLP Findings)* (2026).

C6. Qi, W., **Huang, Fan**, An, J. & Kwak, H. *Is DeepSeek a New Voice Among LLMs in Public Opinion Simulation?* in *Proceedings of the IEEE International Conference on Data Mining Workshops (ICDMW)* (2025).

C7. **Huang, Fan**, Kwak, H., Park, K. & An, J. *ChatGPT Rates Natural Language Explanation Quality like Humans: But on Which Scales?* in *Proceedings of the Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)* (2024).

C8. **Huang, Fan**, Kwak, H. & An, J. *Chain of Explanation: New Prompting Method to Generate Higher Quality Natural Language Explanation for Implicit Hate Speech* in *Companion Proceedings of the Web Conference (WWW)* (2023).

C9. **Huang, Fan**, Kwak, H. & An, J. *Is ChatGPT Better than Human Annotators? Potential and Limitations of ChatGPT in Explaining Implicit Hate Speech* in *Companion Proceedings of the Web Conference (WWW)* (2023).

C10. Liaw, S., **Huang, Fan**, Benevenuto, F., Kwak, H. & An, J. *YouNICon: YouTube's CommuNIty of Conspiracy Videos* in *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)* (2023).

Preprints / Under Review

J1. **Huang, Fan**, Kim, M., Diggans, C. T. & Radicchi, F. Evaluating LLM-Generated Preference Distributions. *arXiv preprint arXiv:2610.01000* (2026).

J2. **Huang, Fan**. Catalyst Papers in Artificial Intelligence Research: A Landscape on ICLR from 2017 to 2025. *arXiv preprint* (2026).

J3. **Huang, Fan**. Reasoning Topology Matters: Network-of-Thought for Complex Reasoning Tasks. *arXiv preprint arXiv:2603.20730* (2026).

J4. **Huang, Fan**. ReFlect: An Effective Harness System for Complex Long-Horizon LLM Reasoning. *arXiv preprint* (2026).

J5. **Huang, Fan**. Simulating Hate Speech Cascades with Multi-LLM Agents: Empirical Grounding, Modeling Fidelity, and Intervention Strategies. *arXiv preprint* (2026).

J6. **Huang, Fan** & Kim, M. Can LLMs Predict Academic Collaboration? Topology Heuristics vs. LLM-Based Link Prediction on Real Co-authorship Networks. *arXiv preprint arXiv:2604.01379* (2026).

J7. **Huang, Fan**, Tong, D., Chen, C., Kwak, H. & An, J. Instruction-Conditioned Motion Control for Robotic Arms: Language-to-Code Decomposition and Reinforcement Learning. *arXiv preprint arXiv:2610.19617* (2026).

J8. **Huang, Fan**, Zhang, S., Kwak, H. & An, J. CogBias: Measuring and Mitigating Cognitive Bias in Large Language Models. *arXiv preprint arXiv:2604.01366* (2026).

J9. **Huang, Fan** & Zhou, H. Comparative Evaluation of LLM-Based Review Paradigms for Academic Paper Quality Assessment. *OpenReview* (2026).

J10. Seckin, O. C., Nascimento Silva, F., Truong, B. T., Kim, S., **Huang, Fan**, Liu, N., Flammini, A. & Menczer, F. The Rise of Bluesky. *arXiv preprint* (2025).

J11. Yin, L. & **Huang, Fan**. DIF: A Framework for Benchmarking and Verifying Implicit Bias in LLMs. *arXiv preprint arXiv:2505.10013* (2025).

J12. **Huang, Fan**, Kwak, H. & An, J. Token-Ensemble Text Generation: On Attacking the Automatic AI-Generated Text Detection. *arXiv preprint* (2024).

Research Impact

Cited by 42 papers that each have 100+ citations, including *Llama 2* (Meta; 18,593 citations) and *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena* (NeurIPS 2023; 12,204 citations). Citation counts from Semantic Scholar (Oct. 2026); each citation below was verified in the citing work's reference list.

Selected Media Coverage

2026 Multi-LLM Agent Simulation Reproduces Hate Speech Cascades with 97% Hostile Stance Accuracy.
The Neural Feed. Jun 18, 2026. Covers J5.

Network-of-Thought (NoT) moves LLM reasoning from linear chains and trees to complex directed graphs, significantly improving multi-hop QA.
SeriesFusion.ai. SeriesFusion Editorial Board. Mar 24, 2026. Covers J3.

How Can We Tell if LLMs Actually Think Like Humans? XCHOICE Offers a Mechanism-Based Answer.
CORe Inc., AI Research. Masato Okuwaki. Jan 22, 2026. Covers C5.

2022 7 Papers & Radios (weekly selection), featuring Chain of Explanation among the week's 10 selected NLP papers.
Synced (Jiqizhixin); syndicated by The Paper and Tencent News. ArXiv Weekly Radiostation. Sep 18, 2022. Covers C8.

Cited in Nature- and PNAS-Family Journals

2026 Evaluating LLMs' divergent thinking capabilities for scientific idea generation with minimal context.
Nature Communications. Kai Ruan et al. Mar 2026. Cites C7.

2024 Perils and opportunities in using large language models in psychological research.
PNAS Nexus. Suhaib Abdurahman et al. Jul 2024. Cites C9.

2023 ChatGPT outperforms crowd workers for text-annotation tasks.
Proceedings of the National Academy of Sciences (PNAS). Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. Jul 2023. Cites C9.

Cited in Other Nature Portfolio Journals

2026 Fallibility, persuadability, and correctability of large language models under sustained conversational misinformation pressure.
Scientific Reports. Jordan Rodriguez et al. Sep 2026. Cites C4.

A Blue Start: A large-scale pairwise and higher-order social network dataset.
Scientific Data. Alyssa Hasegawa Smith et al. Mar 2026. Cites J10.

2025 LLMs outperform outsourced human coders on complex textual analysis.
Scientific Reports. Vicente J. Bermejo et al. Nov 2025. Cites C9.

Machine-assisted quantitizing designs: augmenting humanities and social sciences with artificial intelligence.
Humanities and Social Sciences Communications. Andres Karjus. Feb 2025. Cites C9.

2024 Foundation metrics for evaluating effectiveness of healthcare conversations powered by generative AI.
npj Digital Medicine. Mahyar Abbasian et al. Mar 2024. Cites C8.

Evolving linguistic divergence on polarizing social media.
Humanities and Social Sciences Communications. Andres Karjus and Christine Cuskley. Mar 2024. Cites C9.

Cited by Researchers at Frontier AI Labs

2025 Kurt Thomas et al. (Google and Google DeepMind). May 2025. Cites C9.

2024 Ruibo Liu et al. (Google DeepMind). Oct 2024. Cites C9.

Yifan Wang et al. (Meta). Sep 2024. Cites C9.

Vinodkumar Prabhakaran et al. (Google Research). Jun 2024. Cites C9.

2023 Lora Aroyo et al. (Google Research). Dec 2023. Cites C9.

Hugo Touvron et al. (Meta). Jul 2023. Cites C9.

Christopher M. Homan et al. (Google Research). Jun 2023. Cites C9.

Cited in Government and Policy Reports

2026 Validating Large Language Model Annotations.
Federal Reserve Board, Finance and Economics Discussion Series 2026-020 (staff working paper). Anne Lundgaard Hansen. Mar 2026. Cites C9.

2025 The Role of Artificial Intelligence in Scientific Research: A Science for Policy, European Perspective.
European Commission, Joint Research Centre (JRC), Science for Policy report. Erasmo Purificato et al. Oct 2025. Cites C9.

2024 Technical report on the state of the art of political disinformation in social networks and the press.
HYBRIDS Deliverable D3.2, Horizon Europe Marie Skłodowska-Curie Doctoral Network. HYBRIDS doctoral researchers. Apr 2024. Cites C8.

2023 ChatGPT: research evidence-based controversies, regulations and solutions.
Good In Tech Research Network (Institut Mines-Télécom and Sciences Po), report. Kay Yip and Christine Balagué. Sep 2023. Cites C9.

Awards & Honors

2026 **Google PhD Fellowship - Institutional Nominee**, nominated by Indiana University Bloomington (IUB).

2026 **Travel Fund**, Midwest Speech and Language Days (MSLD) 2026.

2023.01-2027.12 **Full Ph.D. Scholarship**, Indiana University Bloomington (IUB).

2022.01-2022.12 **Full Ph.D. Scholarship**, Singapore Management University (SMU).

2017 **Outstanding Student Leader**, Central South University.

2017 **Second Prize**, Information Security Competition, Central South University.

2017 **Merit Award**, East China Division, China College Student Cloud Computing Competition.

2016 **Second Prize**, U.S. Mathematical Modeling Competition.

2016 **Third Prize**, Shenghua Cup, Central South University.

2016 **Second Prize**, Hunan Province E-Commerce Challenge.

2015 **Shanhe Yingcai Scholarship (Second Class)**.

2015 **First-Class Scholarship (University Level)**, Central South University.

Research Organization

2027 Workshop Proposal, *Delegation, Oversight, and Evaluation of AI Across the Research Lifecycle*, ACM CHI Conference on Human Factors in Computing Systems (CHI 2027). **Huang, Fan**, Yang, K.-C. & Kim, M.

2027 Workshop Proposal, *What Do LLMs Believe? Belief Representation and Human-AI Alignment*, International Conference on Learning Representations (ICLR 2027). **Huang, Fan**, Lee, B., Kwak, H. & An, J.

Teaching

Indiana University Bloomington – Luddy School of Informatics, Computing, and Engineering

2026 Teaching Assistant, INFO-I 201 Mathematical Foundations of Informatics (Fall)

2026 Lecture Presentation, Vision Language Models (VLM), INFO-I 590 Foundation and Application for Large Language Models (Apr)

2026 Lecture Presentation, Retrieval Augmented Generation (RAG), INFO-I 590 Foundation and Application for Large Language Models (Mar)

2026 Teaching Assistant, INFO-I 590 Foundation and Application for Large Language Models (Spring)

2025 Teaching Assistant, INFO-I 590 Foundation and Application for Large Language Models (Fall)

2025 Teaching Assistant, INFO-I 590 Foundation and Application for Large Language Models (Spring)

Research Advising

Master's Student: Sathvik Bandaru (co-advising)
Undergraduate Student: Jack Waren (co-advising)

Research Experience

2023–	Research Assistant, Indiana University Bloomington Advisors: Prof. Jisun An and Prof. Haewoon Kwak Collaborators: Prof. Yong-Yeol Ahn, Prof. Filippo Menczer, and Prof. Filippo Radicchi
2024	Quantitative Research Intern, Core Research Team, QuantLab Financial, Houston (Jun–Aug)
2024	Research Summer Internship, Indiana University Bloomington, supervised by Prof. Katy Börner (May–Jun)
2022–23	Research Assistant, Singapore Management University

Presentations

Talks

T1. **Huang, Fan**. *LLM Belief Atlas.* Poster presentation at InfoVis (Boston). Nov. 2026.
T2. **Huang, Fan**. *Evaluating LLM-Generated Preference Distributions.* Invited research discussion, Squoosh, Inc. Oct. 2026.
T3. **Huang, Fan**. *Trustworthiness of LLM Reasoning Capabilities.* Invited research talk, Prof. Michael Färber's lab, TU Dresden. Sept. 2026.
T4. **Huang, Fan**. *Understanding Moral Reasoning Trajectories in Large Language Models: Toward Probing-Based Explainability.* Parallel talk at IC2S2 (Burlington, Vermont). July 2026.
T5. **Huang, Fan**. *LLM Moral Reasoning Trajectory Analysis.* Invited talk, Prof. Han Xue's lab, Donghua University, Shanghai. June 2026.
T6. **Huang, Fan**. *AI-Assisted Research.* Invited talk, Prof. Thai Le's lab, Indiana University Bloomington. Apr. 2026.
T7. **Huang, Fan**. *Vulnerability of LLMs' Stated Belief? LLMs Belief Resistance Check Through Strategic Persuasive Conversation Interventions.* Presentation at MSLD (UIUC). Mar. 2026.
T8. **Huang, Fan**. *ChatGPT Rates Like Human: Towards a Better Alignment of Text Explanation Quality Assessments.* Oral presentation at MSLD (Ann Arbor, Michigan, USA). 2024.
T9. **Huang, Fan**. *Is ChatGPT Better than Human Annotators? & Chain of Explanation.* Poster presentation at The Web Conference (WWW) (Austin, Texas, USA). 2023.
T10. **Huang, Fan**. *Natural Language Explanation: From XAI to Human-AI Interaction.* YY Group, CNetS, Indiana University Bloomington. 2023.

Workshops

WK1. **Huang, Fan**. *Participant*, *Summer Institute of Computational Social Science (SICSS)*. NUS, Singapore. 2022.
WK2. **Huang, Fan**. *Co-organizer*, *Workshop "Introduction to NLP"*. IEEE NTU Student Branch, NTU, Singapore. Oct. 2021.

Academic Service

Chairs

2026	EMNLP: Virtual Program Session (Topic: Interpretability and Analysis of Models for NLP), Session Chair
2026	IC2S2: Science of Science (AI) Parallel Talk Session, Session Chair

Program Committee Member

2026	WISE
2026	IC2S2
2026	WWW: Web4Good
2026	7th CySoc Workshop at ICWSM
2025	6th CySoc Workshop at ICWSM
2025	WebSci

Conference Reviewer

2027	ICRA
2027	ICLR
2026	NeurIPS
2026	KDD
2026	COLM
2026	IUI
2026	ACL Rolling Review (ARR)
2026	ICWSM
2025	IMX (ACM International Conference on Interactive Media Experiences)
2025	ACL Rolling Review (ARR)
2025	CHI
2025	ICWSM
2025	IDC
2022	WSDM

Journal Reviewer

2025	PeerJ Computer Science
2024–25	Online Social Networks and Media
2024	Information Processing & Management

Organized Activities / Volunteer

2025	SODA Lab Research Talk: 'Multimodal NLP Research Insights', Dr. Henry Wei Han, Organizer (Nov)
Aug 2023–Oct 2024	CNetS @ Luddy School Gathering Event, Organizer
2023	ISSI, Student Volunteer

Campus Leadership and Activities

2024/10–Present	**Founder, LLM Innovators Hub @ CNetS**, Indiana University. Organize LLM research and application sharing sessions within CNetS.
2020/10–2021/10	**Member, IEEE NTU Student Branch**, Nanyang Technological University. Co-organized the workshop "Introduction to NLP".
2015/09–2018/09	**Academic Affairs Representative**, Computer Science Class 1401, Central South University.
2015/09–2017/09	**Director, University Student Union Office**, Central South University.
2014/09–2015/09	**Class Monitor**, Microelectronics Class 1401, Central South University.

Research Committee

Jisun An
Assistant Professor, *Luddy School of Informatics, Computing, and Engineering*,
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Thai Le
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Filippo Radicchi
Professor, *Luddy School of Informatics, Computing, and Engineering*,
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Research Rotation Advisors

Filippo Menczer
Luddy Distinguished Professor of Informatics and Computer Science, *Luddy School of Informatics, Computing, and Engineering*, Indiana University Bloomington
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Yong-Yeol (YY) Ahn
Quantitative Foundation Distinguished Professor of Data Science, *School of Data Science*,
University of Virginia
yyahn@virginia.edu

Last updated: October 30, 2026